BAHAR AYAZI

MASc Student in Electrical and Computer Engineering | Software Engineer

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SUMMARY

Software engineer with 3+ years of experience building reliable backend systems, APIs and data pipelines, now an MASc student in Electrical and Computer Engineering at the University of Victoria. In industry I optimized slow APIs and databases, automated manual financial workflows, and integrated new platforms with legacy systems. In my research and coursework I apply machine learning to anomaly and fraud detection, with a focus on reproducible, well-tested pipelines. I work closely with product, business and data teams to turn requirements into practical solutions.

EDUCATION

MASc, Electrical and Computer Engineering, University of Victoria **Sep 2026 - present**

- Co-supervised by Dr. Ardeshir Shojaeinasab and Dr. Homayoun Najjaran
- Coursework: Data Mining, Firewalls and Intrusion Prevention, Blockchain

BSc, Mechanical Engineering - Biosystems, Isfahan University of Technology **2018 - 2022**

- Relevant coursework: Fundamentals of Computer Engineering(A), Engineering Mathematics(A+), Numerical Methods(A)

RESEARCH & PROJECTS

MASc Research: Security of a Robotic Cell, University of Victoria | Fall 2026 - present

- Working on the security component of a lab robotic cell, collecting data on normal operating behavior to detect abnormal or unsafe behavior.
- Designing the approach to generalize to other systems beyond the robotic cell.

RAFE: Reliability-Aware Fraud Evidence Fusion | Team Lead (4 members) | UVic Data Mining, Fall 2026

Python, scikit-learn, XGBoost, pandas, pytest, uv, GitHub

- Designed a multi-agent fraud detector that splits each transaction into transaction, identity and behavioral feature sub-spaces, scores each with a lightweight agent, and fuses the scores with validation-measured reliability weights; agents with no evidence for a transaction do not vote.
- Built the end-to-end pipeline on 590K IEEE-CIS transactions with a leakage-free chronological split: plug-in agent/fusion interfaces, centralized and fusion baselines, ablation and latency analysis.
- Matched a centralized XGBoost baseline (PR-AUC 0.521 vs 0.517) at 2.8 ms per transaction, and improved PR-AUC by 20% (0.230 vs 0.192) on the 79% of transactions that lack identity data.
- Made every run reproducible (fixed seeds, run records with commit and data hashes, identical results from a fresh clone); led task planning, code reviews and pull-request integration.

Blockchain from Scratch (Python, ECE 519C) | Fall 2026

- Implemented vanity address generation, proof-of-work mining and a 3-node peer-to-peer network.

Firewall Policy Design and Testing (Cisco Packet Tracer, ECE 519B) | Fall 2026

- Built and tested stateless ACLs across Staff, DMZ and Outside zones.

PROFESSIONAL EXPERIENCE

Software Engineer Dotin **Sep 2025 - Jul 2026**

- Developed application-layer services and internal APIs for a middleware platform supporting infrastructure operations in a large-scale banking environment, integrating Kafka, Redis, Consul, ZooKeeper, ActiveMQ and Elasticsearch.
- Investigated legacy architecture and infrastructure tools, identified maintainability and scalability limitations, and refactored components into a modular structure.
- Worked on security-sensitive banking infrastructure involving access control, authentication, auditability and reliability.
- Independently learned unfamiliar infrastructure technologies and built custom components when existing packages did not meet project requirements.
- Translated infrastructure-team requirements and workflow gaps into features; reviewed backend and frontend code and performed end-to-end testing from the perspective of the platform's technical users.

Back-End Developer ActiveCleaners **Nov 2024 - Sep 2025**

- Developed data-driven backend workflows for service estimation, pricing, scheduling, provider assignment and order management on a platform combining online laundry and on-site cleaning.
- Designed logic that combines service type, item characteristics, duration, location, provider availability and capabilities to support estimation and assignment decisions.
- Designed a concurrency-safe shared wallet and payment architecture connecting a new Django application with a legacy .NET application across separate databases.
- Reverse-engineered business rules from the legacy .NET application and reimplemented them in Python/Django while preserving data integrity.
- Worked with product and engineering stakeholders to evaluate alternative solutions, identify edge cases, race conditions and data consistency risks, and design extensible workflows.

Back-End Developer (Remote, Contract) Tradido **Mar 2024 - Nov 2024**

- Developed backend APIs for a real-time cryptocurrency social platform with high-frequency market-data flows and performance-sensitive endpoints.
- Translated product requirements into backend workflows and scalable API designs, balancing performance and user experience.
- Proposed and shipped a lightweight keyword-based search to meet an immediate product need while keeping a migration path to Elasticsearch.
- Contributed to migrating a monolithic application toward microservices in a high-concurrency environment.
- Proactively identified product improvements and implemented new features with stakeholders.

Back-End Developer SnappCarfix **Jan 2023 - Nov 2024**

- Developed and maintained backend services across B2B, B2C, service-center and agent-facing products handling over 1.2 million API requests per day.
- Optimized a product-inventory API whose response time could reach several minutes, using caching, query optimization and targeted SQL to bring it into the millisecond range in production.
- Designed and implemented an automated refund workflow with business-rule validation, payment-gateway integration and concurrency safeguards against duplicate refunds, cutting processing time from about one week to 24-72 hours and removing the manual process through a staged rollout.
- Collaborated with BI specialists on complex SQL and reporting requirements, validating data logic against backend business rules.
- Built asynchronous synchronization services with Celery and improved authentication mechanisms and backend reliability.
- Collaborated with frontend developers and product designers on user flows and API structures.
- Mentored junior backend developers through onboarding, code reviews and technical guidance.

SKILLS

- **Programming & Frameworks:** Python, SQL, Django, Flask, FastAPI
- **Machine Learning & Data:** scikit-learn, XGBoost, Pandas, NumPy
- **Distributed Systems & Messaging:** Kafka, Redis, ZooKeeper, Consul, ActiveMQ, Elasticsearch, Celery
- **Databases:** PostgreSQL, MySQL, MSSQL
- **Infrastructure & DevOps:** Docker, Linux, Git, GitHub, GitLab CI/CD
- **Security & Authentication:** JWT, OAuth2, OpenID Connect (OIDC), LDAP, Cisco ACLs
- **Testing:** Pytest, Performance Testing, End-to-End Testing
- **Soft Skills:** Leadership, Mentoring, Analytical Problem Solving, Cross-functional Collaboration, Translating Business Requirements

LANGUAGE

English: Professional Working Proficiency (CEFR C1 / IELTS 7.0)
Persian: Native